

MUSTAFA GHANIM

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SUMMARY

Computer engineer and PhD candidate with experience spanning semiconductor design, the FPGA and consumer electronics industry, and federally funded research labs. Works across the full stack, from RTL and board bring-up to runtime systems and ML/signal processing applications. Turns open-ended, system-level problems into modular designs, pragmatic, and results-oriented solutions, backed by peer-reviewed publications and an award-winning demonstration. Combines strong technical ownership with the ability to explain complex ideas clearly. Ready to contribute from day one on teams planning, building, and testing next-generation hardware, software, and systems.

TECHNICAL SKILLS

Programming: C,C++, CUDA, Python, MATLAB, Bash	Digital Design: Verilog, SystemVerilog, HLS, TCL
Systems: Slurm, PetaLinux, Docker, Kubernetes, Git	EDA & Simulation: Vivado, Vitis, Verisium, Cadence, Simulink
HW/SW Interfaces: AHB, AXI, PCIe, XDMA, DMA	SoC Platforms: RISC-V/x86/ARM CPUs, NVIDIA GPUs, AMD FPGAs

WORK EXPERIENCE

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|---|---|--|
| Graduate Research Assistant | <i>University of Arizona – Reconfigurable Computing Lab</i> | <i>Aug 2022–Present</i> |
| <ul style="list-style-type: none">Built a lightweight C++ runtime for Domain-Specific SoCs (DSSoCs) that maps and dispatches workloads across heterogeneous CPU, GPU, and FPGA compute units, achieving a 60× reduction in context-switching overhead.Designed HOPPERFISH, a portable C++ real-time profiling framework for anomaly detection on heterogeneous SoCs, with 2.1× lower overhead and an 87.9× smaller memory footprint than vendor tools (Intel VTune, NVIDIA Nsight Systems).Developed multi-node SoC orchestration with containerized execution, integrating a hardware-agnostic CPU/GPU/FPGA scheduler with automated workload-to-hardware mapping (C++, Python), delivering up to 50% faster execution, 43% lower energy, and 1.7× higher throughput compared to a GPU-only platform.Deployed and benchmarked signal processing and ML workloads on FPGA platforms (Alveo U280, VC707, VCU128, ZCU102 MPSoC), using Tcl, Python, and Bash scripts to automate reproducible builds, validate correctness, and plot results.Brought up the Alinx Z19-M (Zynq UltraScale+ MPSoC + Jetson Orin NX), integrating PCIe drivers between both devices and verifying dynamic FFT task mapping across ARM cores, the Orin GPU, and Xilinx FFT IPs. | | |
| Graduate Digital Design Intern | <i>Texas Instruments, Tucson, AZ</i> | <i>May–Aug 2024 & May–Aug 2025</i> |
| <ul style="list-style-type: none">Introduced a Magillem-based flow to automate register-map and HW/SW interface generation for the team’s first ARM-based SoC.Designed a parameterized zero-wait AHB-to-SRAM bridge in RTL; verified it with SystemVerilog testbenches and formal property checks in JasperGold.Developed an AHB-compliant controller for EEPROM and high-voltage generators supporting 10+ operational modes.Simulated transistor-level Single-Event Upsets (SEUs) and implemented RTL mitigation (Hamming coding, majority voting, scrubbing), including an radiation-tolerant memory element for space-grade DAC devices. | | |
| Graduate Teaching Assistant | <i>University of Arizona – ECE Department</i> | <i>Aug 2022–Present</i> |
| <ul style="list-style-type: none">Supported instruction in Introduction to Electromagnetics, Computer Organization, Electronic Circuits, and C Programming; mentored undergraduates and explained complex hardware and software concepts to diverse audiences. | | |
| Graduate Research Assistant | <i>Ozyegin University – nEMESysLab, Istanbul</i> | <i>Sep 2019–Aug 2022</i> |
| <ul style="list-style-type: none">Designed a lossless depth-map compression algorithm exploiting spatial correlations, achieving up to 56% memory reduction compared to SOTA; delivered real-time Verilog RTL implementations for FPGA and ASIC sustaining 1080p @ 60–80 fps.Synthesized the design in SMIC 40 nm (4k–6k gates) with 40% dynamic power reduction compared to SOTA, using gate-level switching-activity analysis for power estimation. | | |
| R&D Intern | <i>AntSis Electronics, Turkiye</i> | <i>Jun–Aug 2019</i> |
| <ul style="list-style-type: none">Implemented a MicroBlaze-based video processing subsystem on an UltraScale+ MPSoC using Vivado Block Design; validated UHD display output via the FMC interface. | | |

EDUCATION

PhD, Electrical and Computer Engineering (GPA: 4.00/4.00)	<i>University of Arizona, USA</i>	<i>Aug 2022–Present</i>
MSc, Electrical and Electronics Engineering (GPA: 4.00/4.00)	<i>Ozyegin University, Turkiye</i>	<i>Sep 2019–Dec 2021</i>
BSc, Electronics and Communication Engineering (GPA: 3.71/4.00)	<i>Istanbul Technical University, Turkiye</i>	<i>Aug 2015–Jul 2019</i>

COLLABORATIVE RESEARCH PROJECTS

DASH-SoC – Domain-Focused Advanced Software-Reconfigurable Heterogeneous SoC	Grant #FA8650-18-2-7860
COCHON – Configurable Optical Communications via Heterogeneous-Processing Optimized Node	Grant #HR00112290043
AZSCI Big Idea Challenge – Space-Based Orbital Data Center Infrastructure	University Grant

SELECTED PUBLICATIONS & ACHIEVEMENTS

- Ghanim, M.** et al., “HOPPERFISH: Holistic Profiling with Portable Extensible and Robust Framework Intended for Systems with Heterogeneity,” *ACM Transactions on Architecture and Code Optimization*, 2025.
- Ghanim, M.** et al., “An Efficient Algorithm for Disparity Map Compression Based on Spatial Correlations and Its Low-Cost Hardware Architecture,” *Integration*, 2023.
- Best Demonstration Award, ECE Graduate Student Research Poster & Demo Symposium, University of Arizona, 2024.
- Peer reviewer: *ACM TACO, Digital Signal Processing, Parallel Computing, Discover Computing*.